Lecture 8 recap

In Lecture 8:

- Learnt a five-step process to calculate power series solutions to ODEs
  1. Substitute power series into ODE
  2. Match powers of t
  3. Make infinite sums over same range
  4. Form recurrence relation
  5. Check number of linearly independent solutions
- Applied method to example problems

Lecture 9

In Lecture 9, we will:

1. Find a power series solution to Bessel's equation
2. Learn the importance of uniform convergence and the Weierstrass test
3. Define *formal* and *genuine* power series solutions

A more challenging problem

Airy differential equation

- Is the 2nd-order, homogeneous linear ODE

$$y''(x) - x\,y(x) = 0$$

- It arises when studying optical phenomena, among other things.
- No solutions in terms of elementary functions.

Search for a power series solution: $y(x) = \sum_{n=0}^{\infty} c_n x^n$

$$\sum_{n=0}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Use $n - 2 \to m + 1$ in first sum

$$\Rightarrow \quad \sum_{m=-3}^{\infty} c_{m+3}(m+3)(m+2)x^{m+1} - \sum_{m=0}^{\infty} c_m x^{m+1} = 0$$

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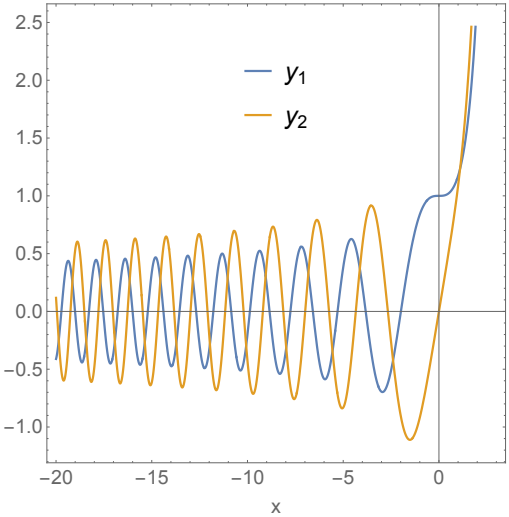
$$c_3 = \frac{1}{2\cdot 3}c_0, \quad \ldots, \quad c_{3k} = \frac{1}{2\cdot 3\cdot 5\cdot 6\cdots(3k-1)\cdot 3k}c_0$$

$$c_4 = \frac{1}{3\cdot 4}c_1, \quad \ldots, \quad c_{3k+1} = \frac{1}{3\cdot 4\cdot 6\cdot 7\cdots 3k\cdot(3k+1)}c_1$$

Two independent solutions:

$$y_1(x) = 1 + \sum_{k=1}^{\infty} \frac{x^{3k}}{2 \cdot 3 \cdot 5 \cdot 6 \cdots (3k-1) \cdot 3k}$$

$$y_2(x) = x + \sum_{k=1}^{\infty} \frac{x^{3k+1}}{3 \cdot 4 \cdot 6 \cdot 7 \cdots 3k \cdot (3k+1)}$$



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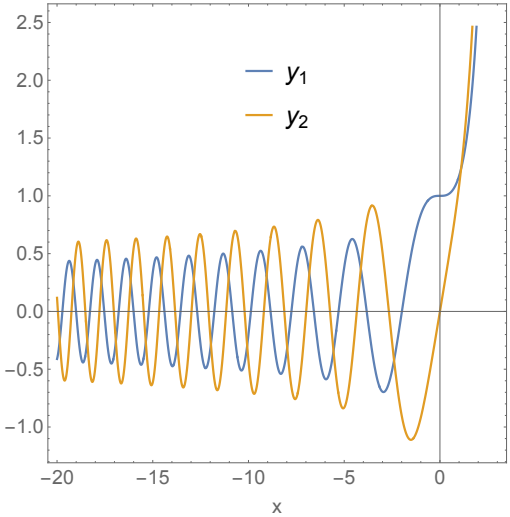

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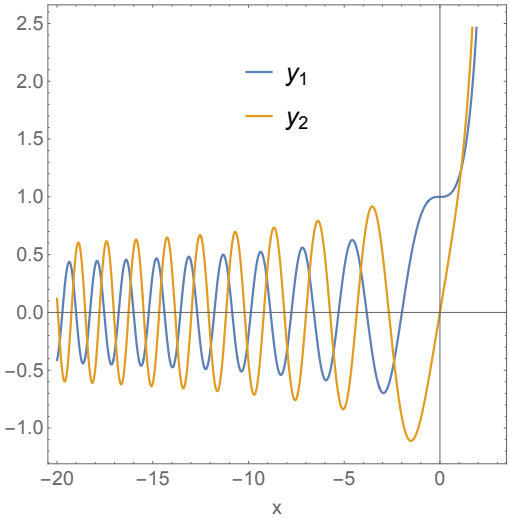
The general solution of Airy's eqn.

$$y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$



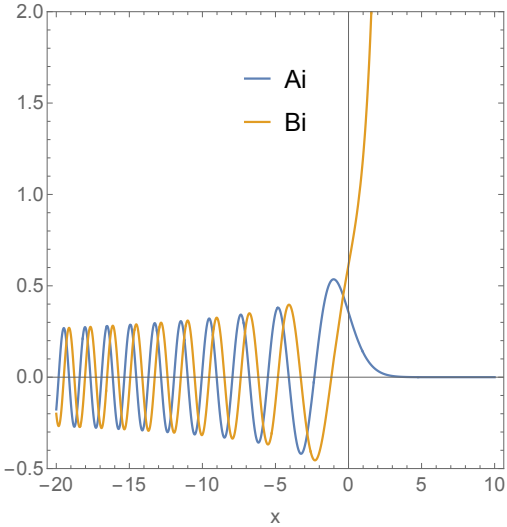
Airy functions

The *Airy functions* are

$$\mathrm{Ai}(x) = \alpha\big[y_1(x) - \beta y_2(x)\big]$$

$$\mathrm{Bi}(x) = \sqrt{3}\alpha\big[y_1(x) + \beta y_2(x)\big]$$

where $\alpha \approx 0.355$ and $\beta \approx 0.729$.



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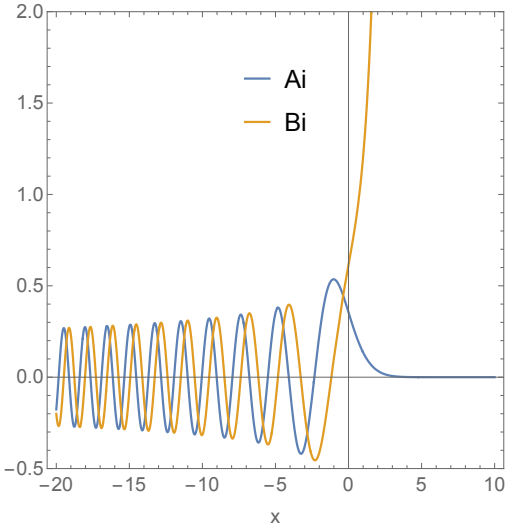

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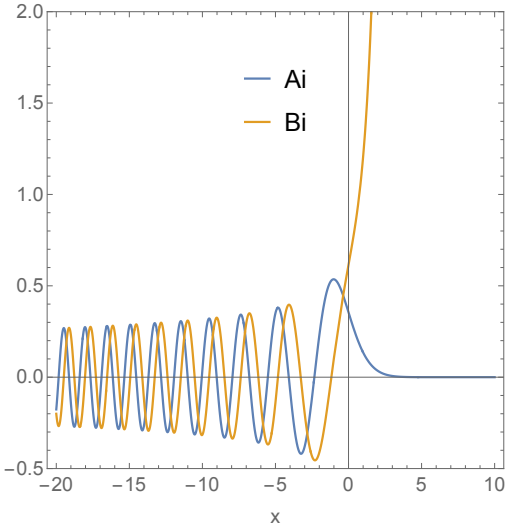

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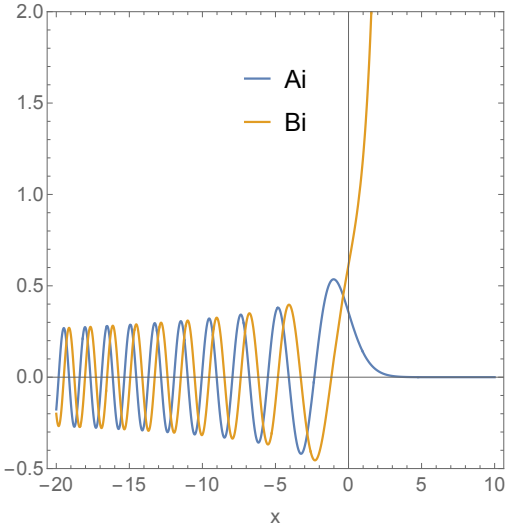
We'll see $\mathrm{Ai}(x)$ again in Topic 6.



Case study 3: Transmission lines and waveguides

- We see electricity transmission or power lines everywhere everyday, but what are waveguides? Well they are basically devices that 'guide' waves, like sound, transmitting them in one direction with minimal loss of energy.
- They are typically hollow and have a rectangular or circular cross section. Modelling the latter leads to a **differential equation that can be solved using the power series method. The solutions are expressed in terms of so-called Bessel's functions.** (Gómez-Correa et al (2017) American Journal of Physics)

Bessel differential equation

Order zero Bessel equation

- Another 2nd-order, homogeneous linear ODE

$$r^2 y''(r) + r y'(r) + r^2 y(r) = 0$$

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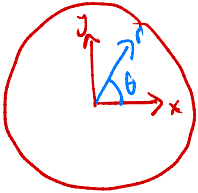

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- ODE <u>does not satisfy</u> requirements for existence and uniqueness of solutions

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$$\Rightarrow \quad \sum_{m=-2}^{\infty} c_{m+2}(m+2)(m+1) r^m + \sum_{m=-2}^{\infty} c_{m+2}(m+2) r^m + \sum_{m=0}^{\infty} c_m r^m = 0$$

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$$\Rightarrow \quad \frac{c_1}{r} + \sum_{m=0}^{\infty} \left\{ c_{m+2}(m+2)^2 + c_m \right\} r^m = 0$$

Therefore, $c_1 = 0$ and $c_{m+2} = -\dfrac{c_m}{(m+2)^2}$ for $m \geqslant 0$.

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The power series method only found one linearly independent solution, which is

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$$y_1(r) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2^k k!)^2} r^{2k} = 1 - \frac{r^2}{4} + \ldots \equiv J_0(r)$$

J_0 is the <u>Bessel function of the first kind of order zero</u>.

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We can't evaluate the integral, and so we just write

$$y(r) = C_1\,J_0(r) + C_2\,J_0(r) \int \frac{1}{rJ_0^2}$$

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$J_0(r) = 1 - r^2/4 + \ldots \approx 1$

for small r

$\int_0^r \frac{1}{\rho J_0^2(\rho)}\,d\rho \approx \int_0^r \frac{1}{\rho}\,d\rho$

$\propto \ln(r)$

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Sequence $W_N(t)$ converges *uniformly* to $W(t)$ on $t \in D$ if

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Stronger condition than pointwise convergence.

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Therefore, $\displaystyle\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converges uniformly on D by the Weierstrass test

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Theorem (Integrating a series)

Then $P(t)$ can be integrated term by term, i.e.,

$$\int_{t_0}^{t} P(\tau)\mathrm{d}\tau = \int_{t_0}^{t} \sum_{n=0}^{\infty} w_n(\tau)\mathrm{d}\tau = \sum_{n=0}^{\infty} \int_{t_0}^{t} w_n(\tau)\mathrm{d}\tau, \qquad t_0, t \in D$$

The integrated series is also uniformly convergent on D.

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- Suppose $P(t)$ converges uniformly on D
- Suppose $\sum_{n=0}^{\infty} w_n'(t)$ converges uniformly on D

Theorem (Differentiating a series)

Then $P(t)$ can be differentiated term by term, i.e.,

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=0}^{\infty} w_n(t) = \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} w_n(t)$$

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Theorem (Uniform convergence of a power series)

Every power series converges uniformly on every closed interval that lies strictly within its interval of convergence.

Formal and genuine solutions

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2. Genuine solution: series and all derivatives up to order of ODE converge uniformly.
   - Term by term differentiation up to the order of the ODE is allowed.
   - Solution satisfies the ODE.

Power series and uniform convergence

Search for a *formal* power series solution to an ODE

$$y(t) = \sum_{n=0}^{\infty} c_n t^n$$

and find coefficients so that the term by term series satisfies the ODE.

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If the radius of convergence is non-zero then, for any domain strictly within the radius of convergence, the power series

1. converges uniformly,
2. can be differentiated term by term and
3. is a *genuine solution* to the ODE.